AI-POWERED DATA MANAGEMENT & MIS REPORTING SYSTEM

(MS Excel, MS PowerPoint & AI Assistance)

October 2026 — Self-Directed Practice Project

Dataset: Sample Business Data (10,200 raw records)

Prepared by

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Self-Directed Project

ACKNOWLEDGEMENT

This is my own self-directed project. It is not a college project, and it was not assigned by any college, university or institution. I planned and completed it on my own to build and show the practical skills used in Data Entry, Back Office and MIS Executive roles.

The dataset used is sample business data, not data from any real company. I used MS Excel for the data work, MS PowerPoint for the management presentation, and an AI assistant to help with checking data quality, standardising entries and drafting insights. Every result was cross-checked against Excel formulas.

I am grateful to my family and friends for their constant support and motivation while I worked on this project. Working on it has helped me apply my classroom knowledge of commerce and spreadsheets to a practical business case, and has strengthened my preparation for a career in data and MIS roles.

Afifah Khatoon

PROJECT OVERVIEW

This project is an end-to-end data management and reporting system. It starts with a messy business dataset of 10,200 raw order records and turns it into a cleaned, validated dataset, an Excel dashboard and a management presentation.

It covers the full office workflow: data preparation, data cleaning, validation, analysis, dashboard reporting and presentation. AI is used as a helper in the analysis stage, while the main data work is done in Excel.

SCOPE OF THE PROJECT

The project covers the following areas:

- Organising and checking a dataset of 10,200 records with 10 fields (Order ID, date, customer, region, category, sales rep, units sold, unit price, total revenue, payment status).
- Finding and fixing duplicates, missing values and inconsistent formats.
- Building an Excel dashboard with key KPIs and pivot-style summary tables.
- Using AI to help find issues, standardise entries and draft insights.
- Preparing an 8-slide management presentation with findings and recommendations.

Out of scope:

- Real company data or live data connections.
- Automated data refresh (Power Query) and Power BI dashboards. These are listed as future work.
- Guessing values that cannot be recovered. Such rows are flagged instead.

PROJECT OBJECTIVES

1. Practise data management, checking and entry rules on a large dataset.
2. Detect and fix data-quality problems: duplicates, blanks, wrong spellings and mixed date formats.
3. Apply validation rules so that the cleaned data is reliable.
4. Build an Excel dashboard that reports revenue, orders, average order value and units sold.
5. Use AI responsibly as an assistant for analysis, with human checking of every result.
6. Present findings and recommendations to management in a clear PowerPoint report.

TOOLS AND KNOWLEDGE USED

Area	What was used
MS Excel	Data formatting, Remove Duplicates, TRIM and PROPER for text cleaning, IF and ISBLANK for missing values, SUMIFS, COUNTIFS and EDATE for summaries, sorting and filtering, charts and dashboard layout
MS PowerPoint	Management presentation with KPI cards, charts, tables and recommendations
AI assistance	Spotting data-quality issues, standardising categories and statuses, checking results, drafting insights and summaries
Knowledge applied	Data validation rules, data cleaning, MIS reporting, basic business analysis, report writing and presentation skills

PROJECT WORKFLOW

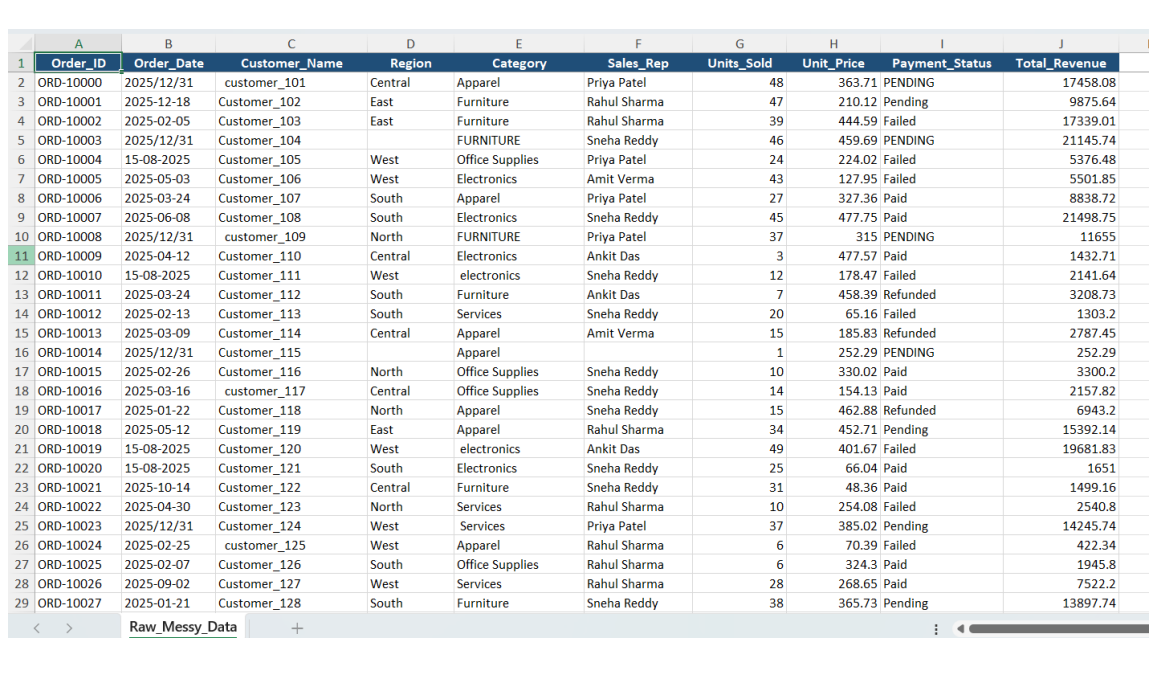
1. **Data preparation:** A raw dataset of 10,200 records with 10 fields was prepared as the starting point.
2. **Data audit:** Duplicates, blank cells, spelling and case differences, and mixed date formats were counted before any change was made.
3. **Data cleaning:** Duplicate rows were removed, text was trimmed and standardised, dates were converted to one real date format, and missing Region and Sales Rep were marked "Unassigned".
4. **Validation:** Total Revenue was checked against Units Sold x Unit Price. Rows that could not be recovered were flagged, not guessed.
5. **Analysis and dashboard:** Summary tables were built with Excel formulas, then turned into KPI cards and charts.
6. **AI-assisted analysis:** AI helped identify issues, suggest standard values and draft insights. Every number was verified in Excel.
7. **Reporting:** Findings and recommendations were presented in an 8-slide PowerPoint deck.

DATASET DESCRIPTION

Field	Description
Order_ID	Unique order number in the format ORD-#####
Order_Date	Date of the order (2025)
Customer_Name	Customer identifier
Region	Central, East, North, South, West
Category	Apparel, Electronics, Furniture, Office Supplies, Services

Field	Description
Sales_Rep	Five sales representatives
Units_Sold	Number of units in the order
Unit_Price	Price per unit
Total_Revenue	Units sold x unit price
Payment_Status	Paid, Pending, Failed, Refunded

Messy Data:



Order_ID	Order_Date	Customer_Name	Region	Category	Sales_Rep	Units_Sold	Unit_Price	Payment_Status	Total_Revenue
ORD-10000	2025/12/31	customer_101	Central	Apparel	Priya Patel	48	363.71	PENDING	17458.08
ORD-10001	2025-12-18	Customer_102	East	Furniture	Rahul Sharma	47	210.12	Pending	9875.64
ORD-10002	2025-02-05	Customer_103	East	Furniture	Rahul Sharma	39	444.59	Failed	17339.01
ORD-10003	2025/12/31	Customer_104		FURNITURE	Sneha Reddy	46	459.69	PENDING	21145.74
ORD-10004	15-08-2025	Customer_105	West	Office Supplies	Priya Patel	24	224.02	Failed	5376.48
ORD-10005	2025-05-03	Customer_106	West	Electronics	Amit Verma	43	127.95	Failed	5501.85
ORD-10006	2025-03-24	Customer_107	South	Apparel	Priya Patel	27	327.36	Paid	8838.72
ORD-10007	2025-06-08	Customer_108	South	Electronics	Sneha Reddy	45	477.75	Paid	21498.75
ORD-10008	2025/12/31	customer_109	North	FURNITURE	Priya Patel	37	315	PENDING	11655
ORD-10009	2025-04-12	Customer_110	Central	Electronics	Ankit Das	3	477.57	Paid	1432.71
ORD-10010	15-08-2025	Customer_111	West	electronics	Sneha Reddy	12	178.47	Failed	2141.64
ORD-10011	2025-03-24	Customer_112	South	Furniture	Ankit Das	7	458.39	Refunded	3208.73
ORD-10012	2025-02-13	Customer_113	South	Services	Sneha Reddy	20	65.16	Failed	1303.2
ORD-10013	2025-03-09	Customer_114	Central	Apparel	Amit Verma	15	185.83	Refunded	2787.45
ORD-10014	2025/12/31	Customer_115		Apparel		1	252.29	PENDING	252.29
ORD-10015	2025-02-26	Customer_116	North	Office Supplies	Sneha Reddy	10	330.02	Paid	3300.2
ORD-10016	2025-03-16	customer_117	Central	Office Supplies	Sneha Reddy	14	154.13	Paid	2157.82
ORD-10017	2025-01-22	Customer_118	North	Apparel	Sneha Reddy	15	462.88	Refunded	6943.2
ORD-10018	2025-05-12	Customer_119	East	Apparel	Rahul Sharma	34	452.71	Pending	15392.14
ORD-10019	15-08-2025	Customer_120	West	electronics	Ankit Das	49	401.67	Failed	19681.83
ORD-10020	15-08-2025	Customer_121	South	Electronics	Sneha Reddy	25	66.04	Paid	1651
ORD-10021	2025-10-14	Customer_122	Central	Furniture	Sneha Reddy	31	48.36	Paid	1499.16
ORD-10022	2025-04-30	Customer_123	North	Services	Rahul Sharma	10	254.08	Failed	2540.8
ORD-10023	2025/12/31	Customer_124	West	Services	Priya Patel	37	385.02	Pending	14245.74
ORD-10024	2025-02-25	customer_125	West	Apparel	Rahul Sharma	6	70.39	Failed	422.34
ORD-10025	2025-02-07	Customer_126	South	Office Supplies	Rahul Sharma	6	324.3	Paid	1945.8
ORD-10026	2025-09-02	Customer_127	West	Services	Rahul Sharma	28	268.65	Paid	7522.2
ORD-10027	2025-01-21	Customer_128	South	Furniture	Sneha Reddy	38	365.73	Pending	13897.74

Messy Data File Link:

https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Raw_Messy_Business_Data.xlsx

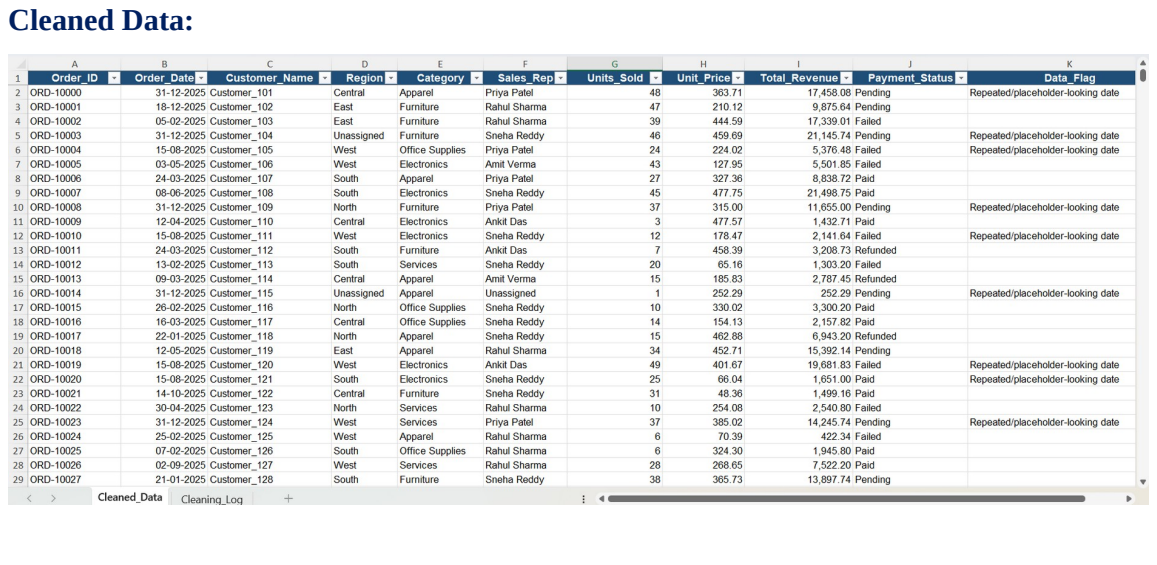
DATA QUALITY PROBLEMS AND CLEANING

The duplicate count is from the 10,200 raw rows. All other counts are on the 10,000 rows left after removing the duplicates.

Issue found	Count	Action taken
Duplicate rows (in raw data)	200	Removed, first copy kept
Mixed date formats	1,457	Converted to real dates, flagged
Inconsistent text (category, status, customer)	3,550 cells	Trimmed spaces, fixed case
Missing Region	500	Marked "Unassigned"
Missing Sales Rep	400	Marked "Unassigned"
Missing Price and Revenue	300	Not guessed, flagged for follow-up
Missing Order ID	197	Not guessed, flagged for follow-up

The 1,457 mixed-format dates are two repeated values (15-08-2025 and 31-12-2025). They look like placeholders, so they were kept and flagged. After cleaning, 10,000 rows remain, and 1,627 rows carry a flag for follow-up. The flags overlap (for example, all 300 rows with missing price and revenue also carry one of the two repeated dates), so the separate issue counts add up to more than 1,627.

Cleaned Data:

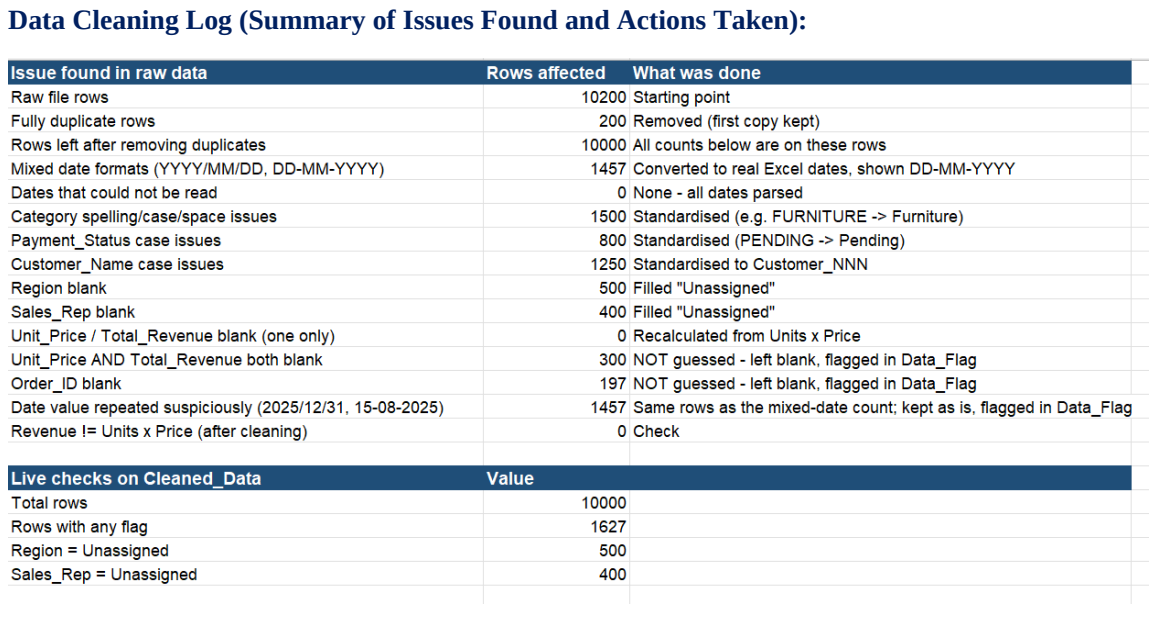


Order_ID	Order_Date	Customer_Name	Region	Category	Sales_Rep	Units_Sold	Unit_Price	Total_Revenue	Payment_Status	Data_Flag
ORD-10000	31-12-2025	Customer_101	Central	Apparel	Priya Patel	48	363.71	17,458.08	Pending	Repeated/placeholder-looking date
ORD-10001	18-12-2025	Customer_102	East	Furniture	Rahul Sharma	47	210.12	9,875.64	Pending	
ORD-10002	05-02-2025	Customer_103	East	Furniture	Rahul Sharma	39	444.59	17,339.01	Failed	
ORD-10003	31-12-2025	Customer_104	Unassigned	Furniture	Sneha Reddy	46	459.69	21,145.74	Pending	Repeated/placeholder-looking date
ORD-10004	15-08-2025	Customer_105	West	Office Supplies	Priya Patel	24	224.02	5,376.48	Failed	Repeated/placeholder-looking date
ORD-10005	03-05-2025	Customer_106	West	Electronics	Amit Verma	43	127.95	5,501.85	Failed	
ORD-10006	24-03-2025	Customer_107	South	Apparel	Priya Patel	27	327.36	8,838.72	Paid	
ORD-10007	08-06-2025	Customer_108	South	Electronics	Sneha Reddy	45	477.75	21,498.75	Paid	
ORD-10008	31-12-2025	Customer_109	North	Furniture	Priya Patel	37	315.00	11,655.00	Pending	Repeated/placeholder-looking date
ORD-10009	12-04-2025	Customer_110	Central	Electronics	Ankit Das	3	477.57	1,432.71	Paid	
ORD-10010	15-08-2025	Customer_111	West	Electronics	Sneha Reddy	12	178.47	2,141.64	Failed	Repeated/placeholder-looking date
ORD-10011	24-03-2025	Customer_112	South	Furniture	Ankit Das	7	458.39	3,208.73	Refunded	
ORD-10012	13-02-2025	Customer_113	South	Services	Sneha Reddy	20	65.16	1,303.20	Failed	
ORD-10013	09-03-2025	Customer_114	Central	Apparel	Amit Verma	15	185.83	2,787.45	Refunded	
ORD-10014	31-12-2025	Customer_115	Unassigned	Apparel	Unassigned	1	252.29	252.29	Pending	Repeated/placeholder-looking date
ORD-10015	26-02-2025	Customer_116	North	Office Supplies	Sneha Reddy	10	330.02	3,300.20	Paid	
ORD-10016	16-03-2025	Customer_117	Central	Office Supplies	Sneha Reddy	14	154.13	2,157.82	Paid	
ORD-10017	22-01-2025	Customer_118	North	Apparel	Sneha Reddy	15	462.88	6,943.20	Refunded	
ORD-10018	12-05-2025	Customer_119	East	Apparel	Rahul Sharma	34	452.71	15,392.14	Pending	
ORD-10019	15-08-2025	Customer_120	West	Electronics	Ankit Das	49	401.67	19,681.83	Failed	Repeated/placeholder-looking date
ORD-10020	15-08-2025	Customer_121	South	Electronics	Sneha Reddy	25	66.04	1,651.00	Paid	Repeated/placeholder-looking date
ORD-10021	14-10-2025	Customer_122	Central	Furniture	Sneha Reddy	31	48.36	1,499.16	Paid	
ORD-10022	30-04-2025	Customer_123	North	Services	Rahul Sharma	10	254.08	2,540.80	Failed	
ORD-10023	31-12-2025	Customer_124	West	Services	Priya Patel	37	385.02	14,245.74	Pending	Repeated/placeholder-looking date
ORD-10024	25-02-2025	Customer_125	West	Apparel	Rahul Sharma	6	70.39	422.34	Failed	
ORD-10025	07-02-2025	Customer_126	South	Office Supplies	Rahul Sharma	6	324.30	1,945.80	Paid	
ORD-10026	02-09-2025	Customer_127	West	Services	Rahul Sharma	28	268.65	7,522.20	Paid	
ORD-10027	21-01-2025	Customer_128	South	Furniture	Sneha Reddy	38	365.73	13,897.74	Pending	

Explanation:

- **Removed duplicates:** Deleted 200 fully duplicate rows and kept the first copy. In Excel: Data > Remove Duplicates.
- **Fixed text:** Removed extra spaces and made the capitalisation consistent (FURNITURE to Furniture, PENDING to Pending, and customer names too). In Excel: =TRIM(PROPER()).
- **Standardised dates:** Converted three different formats (2025/12/31, 2025-12-18, 15-08-2025) into real Excel dates, shown as DD-MM-YYYY (as in column B of the Cleaned_Data sheet). In Excel: DATEVALUE or Text to Columns.
- **Filled blank Region and Sales_Rep:** Marked them "Unassigned" (visible in rows 5 and 16 of the Cleaned_Data sheet). In Excel: =IF(ISBLANK(D2),"Unassigned",D2).
- **Checked revenue:** Verified that Units × Unit_Price = Total_Revenue on every row, with 0 mismatches. In Excel: a helper column with =G2*H2-I2.
- **Did not guess unrecoverable values:** 300 rows had both Price and Revenue blank, and 197 rows had a blank Order_ID. These were left blank and marked in the Data_Flag column.
- **Flagged suspicious dates:** 31-12-2025 and 15-08-2025 repeat very often and look like placeholders. The dates were kept, but Data_Flag says "Repeated/placeholder-looking date" (column K).
- **Recorded everything in the Cleaning_Log sheet:** It lists each issue, how many rows it affected and what was done.

Data Cleaning Log (Summary of Issues Found and Actions Taken):



Issue found in raw data	Rows affected	What was done
Raw file rows	10200	Starting point
Fully duplicate rows	200	Removed (first copy kept)
Rows left after removing duplicates	10000	All counts below are on these rows
Mixed date formats (YYYY/MM/DD, DD-MM-YYYY)	1457	Converted to real Excel dates, shown DD-MM-YYYY
Dates that could not be read	0	None - all dates parsed
Category spelling/case/space issues	1500	Standardised (e.g. FURNITURE -> Furniture)
Payment_Status case issues	800	Standardised (PENDING -> Pending)
Customer_Name case issues	1250	Standardised to Customer_NNN
Region blank	500	Filled "Unassigned"
Sales_Rep blank	400	Filled "Unassigned"
Unit_Price / Total_Revenue blank (one only)	0	Recalculated from Units x Price
Unit_Price AND Total_Revenue both blank	300	NOT guessed - left blank, flagged in Data_Flag
Order_ID blank	197	NOT guessed - left blank, flagged in Data_Flag
Date value repeated suspiciously (2025/12/31, 15-08-2025)	1457	Same rows as the mixed-date count; kept as is, flagged in Data_Flag
Revenue != Units x Price (after cleaning)	0	Check

Live checks on Cleaned_Data	Value
Total rows	10000
Rows with any flag	1627
Region = Unassigned	500
Sales_Rep = Unassigned	400

Explanation:

What this sheet is

- It is the record of the cleaning work. It lists each problem found in the raw data, how many rows it affected and what was done about it.
- The first block is the cleaning summary. The second block, "Live checks on Cleaned_Data", is a set of formulas that recount the cleaned data, so you can confirm the results at any time.

Cleaning summary (first block)

- **Row counts:** the raw file had 10,200 rows. After removing 200 duplicates, 10,000 rows were left. All counts below that line are on these 10,000 rows.
- **Mixed date formats (1,457):** the dates came in different formats and were converted to real Excel dates, shown as DD-MM-YYYY. No date was left unreadable (0).
- **Text issues:** 1,500 category entries, 800 payment status entries and 1,250 customer names had wrong spelling, case or extra spaces. They were standardised (for example FURNITURE to Furniture and PENDING to Pending).
- **Blank Region (500) and Sales_Rep (400):** these were filled with "Unassigned" so that no rows are lost.
- **Blank price or revenue, one only (0):** there were no rows where only one of them was missing, so nothing could be recalculated.
- **Price and revenue both blank (300) and Order_ID blank (197):** these values cannot be recovered, so they were not guessed. They were left blank and flagged in the Data_Flag column.
- **Repeated suspicious dates (1,457):** the dates 2025/12/31 and 15-08-2025 appear too often and look like placeholders. They are the same rows as the mixed-date count, so they were kept and flagged.
- **Revenue check (0):** Total_Revenue equals Units × Price in every row that has a price, so there are 0 mismatches.

Live checks (second block)

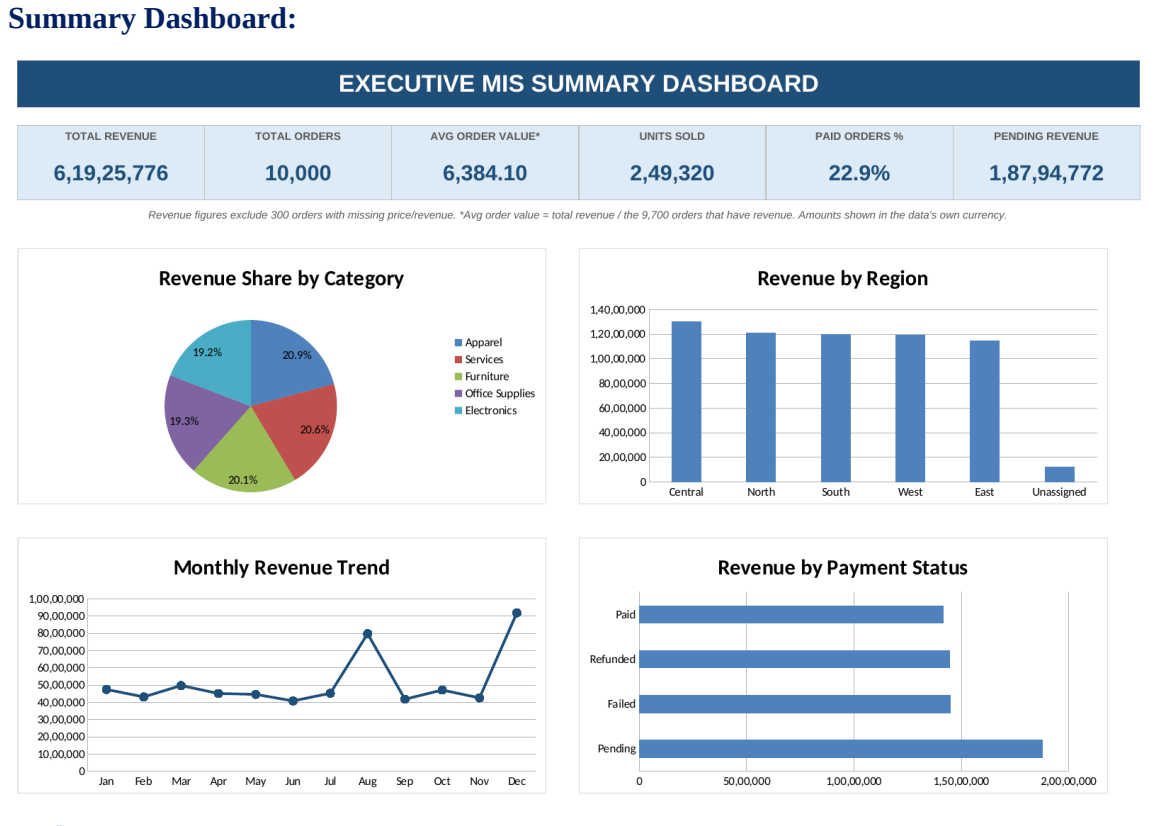
- **Total rows (10,000):** matches the cleaned dataset and the dashboard's total orders.
- **Rows with any flag (1,627):** the number of rows that need follow-up because of a missing value or a suspicious date. One row can have more than one flag, so this is less than the sum of the separate issue counts.
- **Region = Unassigned (500) and Sales_Rep = Unassigned (400):** these match the blank counts above.

DASHBOARD AND KEY RESULTS

KPI	Result
Total revenue	6.19 crore (6,19,25,776)
Total orders	10,000
Average order value	6,384.10
Units sold	249,320
Paid orders	22.9%
Pending revenue	1.88 crore (1,87,94,772)

Revenue figures exclude the 300 orders with missing price and revenue, and the average order value is calculated on the 9,700 orders that have revenue. The dashboard includes charts for revenue by category, revenue by region, monthly trend and payment status. All numbers come from live Excel formulas.

Summary Dashboard:



EXECUTIVE MIS SUMMARY DASHBOARD

TOTAL REVENUE	TOTAL ORDERS	AVG ORDER VALUE*	UNITS SOLD	PAID ORDERS %	PENDING REVENUE
6,19,25,776	10,000	6,384.10	2,49,320	22.9%	1,87,94,772

Revenue figures exclude 300 orders with missing price/revenue. *Avg order value = total revenue / the 9,700 orders that have revenue. Amounts shown in the data's own currency.

Explanation:

KPI cards (top row)

- **Total Revenue, 6,19,25,776:** the sum of Total_Revenue across all orders. In Indian number format this is about 6.19 crore.
- **Total Orders, 10,000:** the number of cleaned order rows, including 300 orders that have no revenue.
- **Avg Order Value, 6,384.10:** total revenue divided by the 9,700 orders that have revenue, not by all 10,000 orders.
- **Units Sold, 2,49,320:** the total units across all orders.
- **Paid Orders %, 22.9%:** only about 1 in 4 orders is fully paid.
- **Pending Revenue, 1,87,94,772:** about 1.88 crore is still waiting to be collected.
- **Note under the cards:** revenue figures exclude the 300 orders with missing price and revenue, and the average order value (marked with *) uses the 9,700 orders that have revenue.

Charts

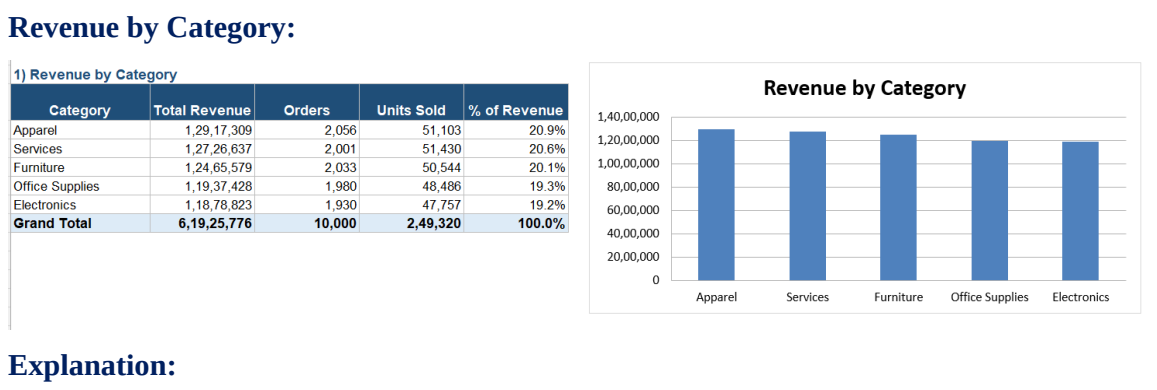
- **Revenue Share by Category (pie):** all five categories are close to 20% each. Apparel leads at 20.9%, and Electronics is lowest at 19.2%. Revenue is evenly spread, with no single dominant category.
- **Revenue by Region (bar):** Central is the highest. North, South, West and East are close together. The small "Unassigned" bar is 500 orders with no region, which points to a data entry gap.
- **Monthly Revenue Trend (line):** most months sit around 40 to 50 lakh. August and December spike sharply. These spikes come from the placeholder dates (15-08-2025 and 31-12-2025), so they are a data quality problem and not real growth.
- **Revenue by Payment Status (bar):** Pending is the largest bar at 30.4% of revenue, while Paid is the smallest at 22.9%.

How it is built

- Every number comes from live Excel formulas (SUMIFS, COUNTIFS) on the Cleaned_Data sheet, linked through the Pivot_Tables sheet. If the data changes, the dashboard updates itself.

Pivot-Style Summary Tables & Charts:

Revenue by Category:



1) Revenue by Category

Category	Total Revenue	Orders	Units Sold	% of Revenue
Apparel	1,29,17,309	2,056	51,103	20.9%
Services	1,27,26,637	2,001	51,430	20.6%
Furniture	1,24,65,579	2,033	50,544	20.1%
Office Supplies	1,19,37,428	1,980	48,486	19.3%
Electronics	1,18,78,823	1,930	47,757	19.2%
Grand Total	6,19,25,776	10,000	2,49,320	100.0%

Explanation:

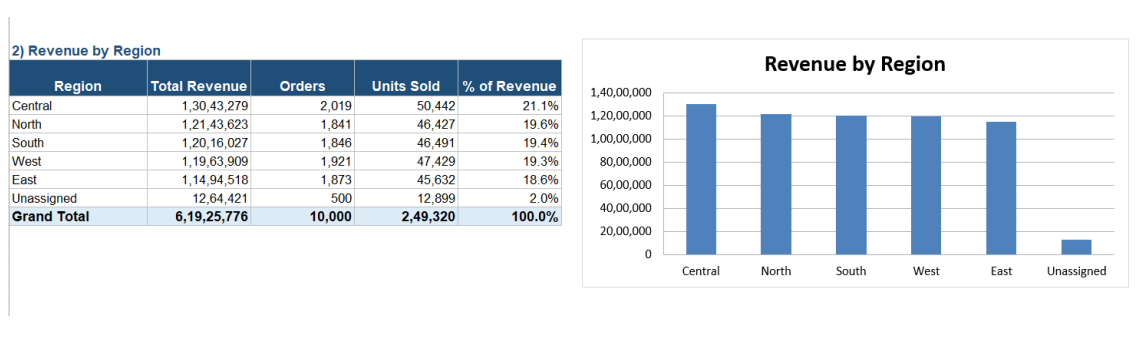
The table: Revenue by Category

- Category: the five product groups: Apparel, Services, Furniture, Office Supplies and Electronics.
- Total Revenue: the revenue each category earned. Apparel is highest at 1,29,17,309 and Electronics is lowest at 1,18,78,823.
- Orders: the number of orders in each category. Apparel has the most (2,056) and Electronics the fewest (1,930).
- Units Sold: the total units sold. Apparel sold 51,103 units and Electronics 47,757.
- % of Revenue: each category's share of total revenue. Apparel has 20.9% and Electronics 19.2%.
- Grand Total: 6,19,25,776 revenue, 10,000 orders, 2,49,320 units and 100%. These match the KPI cards on the dashboard.

The chart: Revenue by Category

- It is a column chart of the same Total Revenue column, with one bar per category.
- The bars are sorted from highest to lowest, so Apparel is on the left and Electronics on the right.
- The vertical axis starts at 0, so the heights are an honest comparison.
- All five bars are almost the same height. Revenue is spread evenly across categories.

Revenue by Region:



2) Revenue by Region

Region	Total Revenue	Orders	Units Sold	% of Revenue
Central	1,30,43,279	2,019	50,442	21.1%
North	1,21,43,623	1,841	46,427	19.6%
South	1,20,16,027	1,846	46,491	19.4%
West	1,19,63,909	1,921	47,429	19.3%
East	1,14,94,518	1,873	45,632	18.6%
Unassigned	12,64,421	500	12,899	2.0%
Grand Total	6,19,25,776	10,000	2,49,320	100.0%

Explanation:

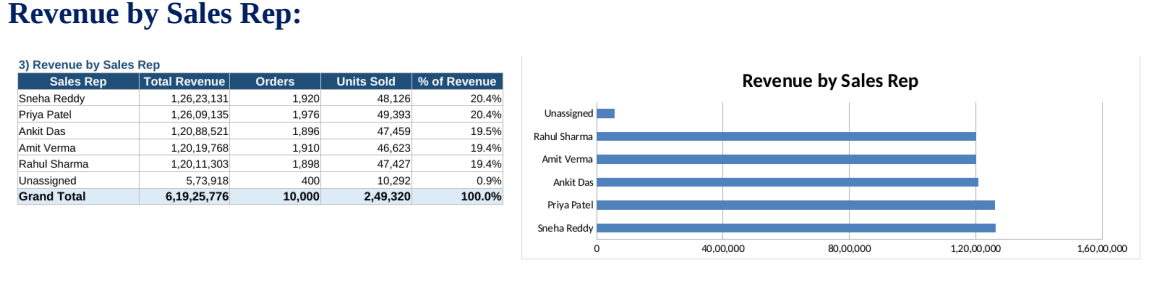
The table: Revenue by Region

- Region: the five sales regions (Central, North, South, West, East) plus "Unassigned" for orders that had no region.
- Total Revenue: Central is highest at 1,30,43,279. East is lowest of the five named regions at 1,14,94,518.
- Orders: Central has the most orders (2,019). East has 1,873 and North has 1,841.
- Units Sold: Central sold the most units (50,442). East sold 45,632.
- % of Revenue: Central has 21.1%. North, South and West are close to each other at about 19.3% to 19.6%. East has 18.6%.
- Unassigned row: 500 orders had a blank region in the raw data. They are shown as their own row so that the Grand Total stays complete.
- Grand Total: 6,19,25,776 revenue, 10,000 orders, 2,49,320 units and 100%. These match the dashboard KPI cards and the Category table.

The chart: Revenue by Region

- It is a column chart of the Total Revenue column, with one bar per region, sorted from highest to lowest.
- The vertical axis starts at 0, so the bar heights are an honest comparison.
- The five named regions have similar bars, with Central slightly taller.
- The "Unassigned" bar is very short. The reason is that 300 of its 500 orders have no price or revenue at all. Only 200 have revenue. So the short bar is a data gap and does not mean low sales.

Revenue by Sales Rep:



3) Revenue by Sales Rep

Sales Rep	Total Revenue	Orders	Units Sold	% of Revenue
Sneha Reddy	1,26,23,131	1,920	48,126	20.4%
Priya Patel	1,26,09,135	1,976	49,393	20.4%
Ankit Das	1,20,88,521	1,896	47,459	19.5%
Amit Verma	1,20,19,768	1,910	46,623	19.4%
Rahul Sharma	1,20,11,303	1,898	47,427	19.4%
Unassigned	5,73,918	400	10,292	0.9%
Grand Total	6,19,25,776	10,000	2,49,320	100.0%

Explanation:

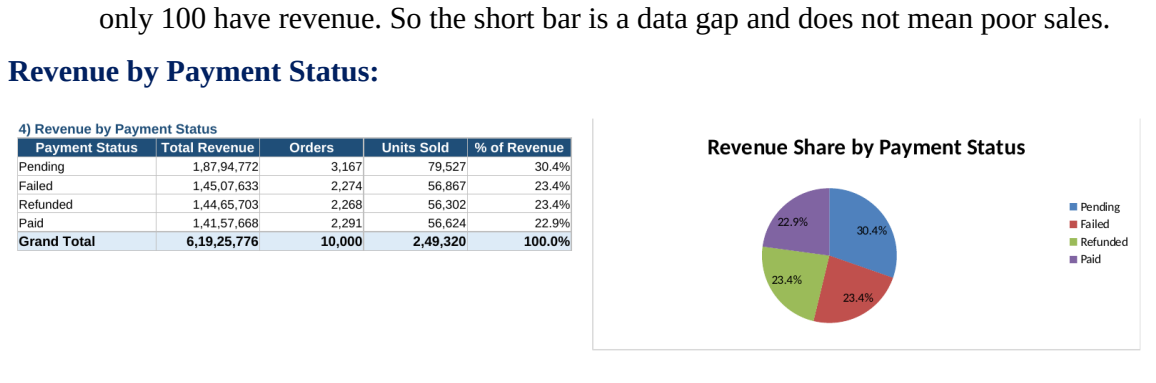
The table: Revenue by Sales Rep

- Sales Rep: the five sales representatives (Sneha Reddy, Priya Patel, Ankit Das, Amit Verma, Rahul Sharma) plus "Unassigned" for orders with no rep name.
- Total Revenue: Sneha Reddy is highest at 1,26,23,131. Priya Patel is almost the same at 1,26,09,135. The other three are close behind at about 1.20 crore.
- Orders: Priya Patel handled the most orders (1,976). Sneha Reddy handled 1,920.
- Units Sold: Priya Patel also sold the most units (49,393).
- % of Revenue: Sneha Reddy and Priya Patel have 20.4% each. Ankit Das, Amit Verma and Rahul Sharma have 19.4% to 19.5%.
- Unassigned row: 400 orders had a blank sales rep in the raw data, so they are kept in their own row. This keeps the Grand Total complete.
- Grand Total: 6,19,25,776 revenue, 10,000 orders, 2,49,320 units and 100%. These match the dashboard KPI cards and the other pivot tables.

The chart: Revenue by Sales Rep

- It is a horizontal bar chart of the Total Revenue column, with one bar per rep.
- The axis starts at 0, so the bar lengths are an honest comparison.
- The five named reps have almost equal bars. Sneha Reddy and Priya Patel are slightly longer.
- The "Unassigned" bar is very short. Of its 400 orders, 300 have no price or revenue, and only 100 have revenue. So the short bar is a data gap and does not mean poor sales.

Revenue by Payment Status:



4) Revenue by Payment Status

Payment Status	Total Revenue	Orders	Units Sold	% of Revenue
Pending	1,87,94,772	3,167	79,527	30.4%
Failed	1,45,07,633	2,274	56,867	23.4%
Refunded	1,44,65,703	2,268	56,302	23.4%
Paid	1,41,57,668	2,291	56,624	22.9%
Grand Total	6,19,25,776	10,000	2,49,320	100.0%

Explanation:

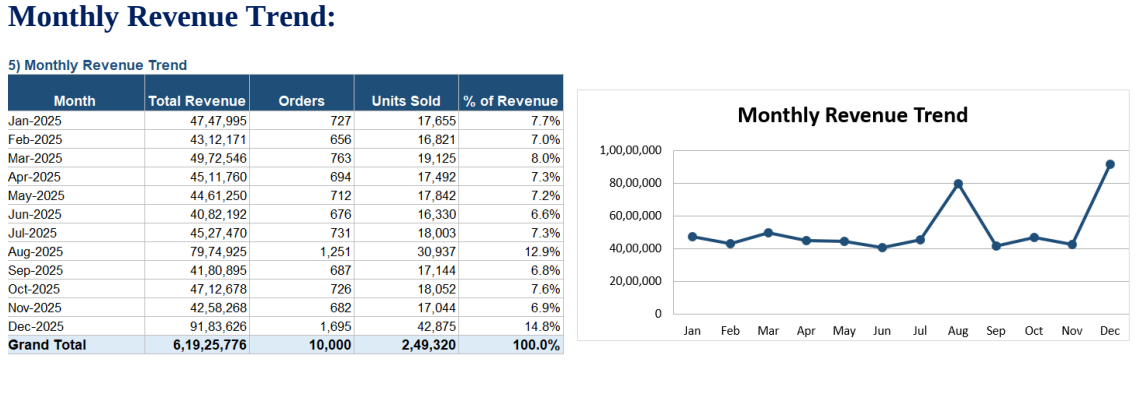
The table: Revenue by Payment Status

- Payment Status: the four statuses an order can have: Pending, Failed, Refunded and Paid.
- Total Revenue: Pending is the biggest at 1,87,94,772. Failed (1,45,07,633) and Refunded (1,44,65,703) are almost equal. Paid is the smallest at 1,41,57,668.
- Orders: Pending has 3,167 orders. The other three have between 2,268 and 2,291 orders each.
- Units Sold: Pending also has the most units (79,527). The other statuses have about 56,000 to 57,000 units.
- % of Revenue: Pending 30.4%, Failed 23.4%, Refunded 23.4% and Paid 22.9%.
- Grand Total: 6,19,25,776 revenue, 10,000 orders, 2,49,320 units and 100%. These match the dashboard KPI cards and the other pivot tables.

The chart: Revenue Share by Payment Status

- It is a pie chart of the same Total Revenue column. Each slice shows the share of one status.
- The Pending slice (blue) is the largest. The other three slices are almost equal.
- The pie labels show one decimal (30.4%, 23.4%, 23.4% and 22.9%), so they match the table exactly.

Monthly Revenue Trend:



5) Monthly Revenue Trend

Month	Total Revenue	Orders	Units Sold	% of Revenue
Jan-2025	47,47,995	727	17,655	7.7%
Feb-2025	43,12,171	656	16,821	7.0%
Mar-2025	49,72,546	763	19,125	8.0%
Apr-2025	45,11,760	694	17,492	7.3%
May-2025	44,61,250	712	17,842	7.2%
Jun-2025	40,82,192	676	16,330	6.6%
Jul-2025	45,27,470	731	18,003	7.3%
Aug-2025	79,74,925	1,251	30,937	12.9%
Sep-2025	41,80,895	687	17,144	6.8%
Oct-2025	47,12,678	726	18,052	7.6%
Nov-2025	42,58,268	682	17,044	6.9%
Dec-2025	91,83,626	1,695	42,875	14.8%
Grand Total	6,19,25,776	10,000	2,49,320	100.0%

Explanation:

The table: Monthly Revenue Trend

- Month: all 12 months of 2025, from Jan-2025 to Dec-2025.
- Total Revenue: most months sit between 40 lakh and 50 lakh. March is the highest normal month at 49,72,546 and June is the lowest at 40,82,192.
- Orders: most months have 650 to 765 orders. August has 1,251 and December has 1,695, which is far above normal.
- Units Sold: it follows the orders. Most months have 16,000 to 19,000 units, while August (30,937) and December (42,875) are much higher.
- % of Revenue: most months are 6.6% to 8.0%. August is 12.9% and December is 14.8%.
- Grand Total: 6,19,25,776 revenue, 10,000 orders and 2,49,320 units. These match the dashboard KPI cards and the other pivot tables.

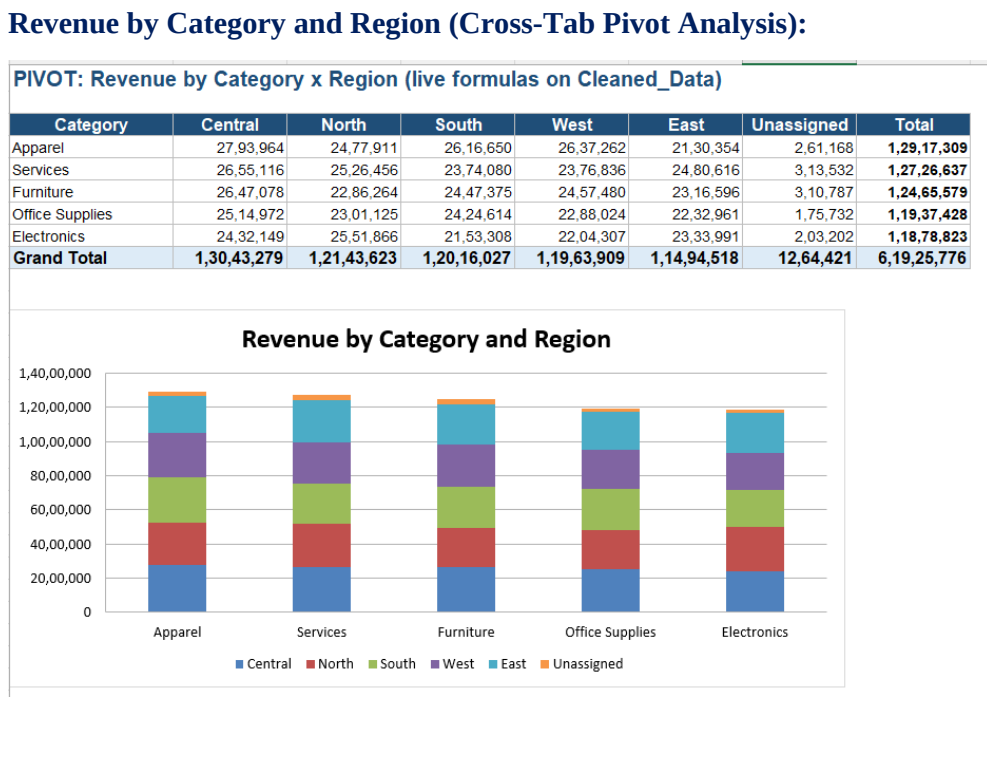
The chart: Monthly Revenue Trend

- It is a line chart of the Total Revenue column, with one point per month.
- The line stays almost flat from January to July, then jumps in August, drops back in September, and jumps again in December.
- The vertical axis starts at 0, so the heights are an honest comparison.

Important: the two spikes are not real growth

- Aug-2025: 521 orders (worth 31,10,196) carry the date 15-08-2025, a single day. The other days of August look like a normal month, about 48.6 lakh.
- Dec-2025: 984 orders (worth 46,80,495) carry the date 31-12-2025, a single day. The rest of December is about 45 lakh, which is normal.
- These two dates repeat far too often to be real, so they look like placeholder dates entered when the real date was missing or unreadable. They are flagged in the Data_Flag column. Of the 1,505 orders on these two days (521 + 984), 1,457 carry the repeated dates and the other 48 are normal orders placed on those days.
- Without these two dates, every month is roughly 40 to 50 lakh, so sales are fairly steady across the year.

Revenue by Category and Region (Cross-Tab Pivot Analysis):



PIVOT: Revenue by Category x Region (live formulas on Cleaned_Data)

Category	Central	North	South	West	East	Unassigned	Total
Apparel	27,93,964	24,77,911	26,16,650	26,37,262	21,30,354	2,61,168	1,29,17,309
Services	26,55,116	25,26,456	23,74,080	23,76,836	24,80,616	3,13,532	1,27,26,637
Furniture	26,47,078	22,86,264	24,47,375	24,57,480	23,16,596	3,10,787	1,24,65,579
Office Supplies	25,14,972	23,01,125	24,24,614	22,88,024	22,32,961	1,75,732	1,19,37,428
Electronics	24,32,149	25,51,866	21,53,308	22,04,307	23,33,991	2,03,202	1,18,78,823
Grand Total	1,30,43,279	1,21,43,623	1,20,16,027	1,19,63,909	1,14,94,518	12,64,421	6,19,25,776

Explanation:

The table

- Highest cell: Apparel in Central, at 27,93,964.
- Lowest named cell: Apparel in East, at 21,30,354. So Apparel is the best seller in Central but the weakest in East.
- Electronics: it has the lowest total (1,18,78,823), but it is the strongest category in North (25,51,866).
- Services: it is steady across regions, with every named region between 23.7 lakh and 26.6 lakh.
- Central: it is the top region overall (1,30,43,279), and it leads in four of the five categories (all except Electronics, where North is ahead).
- Unassigned column: these are orders with no region (12,64,421 in total). They are kept so that the totals stay complete.

The chart

- It is a stacked column chart. Each column is one category, and the coloured parts are the regions (Central, North, South, West, East, Unassigned).
- The five columns are almost the same height, so category totals are close, with Apparel on top and Electronics at the bottom.
- The coloured parts in each column are also similar in size, so no region depends on one category.
- The thin orange part at the top of each column is "Unassigned". It is small because 300 of its 500 orders have no revenue.
- The vertical axis starts at 0, so the heights are an honest comparison.

Excel File Link:

https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/Executive%20MIS%20Dashboard.xlsx

AI-Assisted Management Report in PowerPoint (8 Slides):

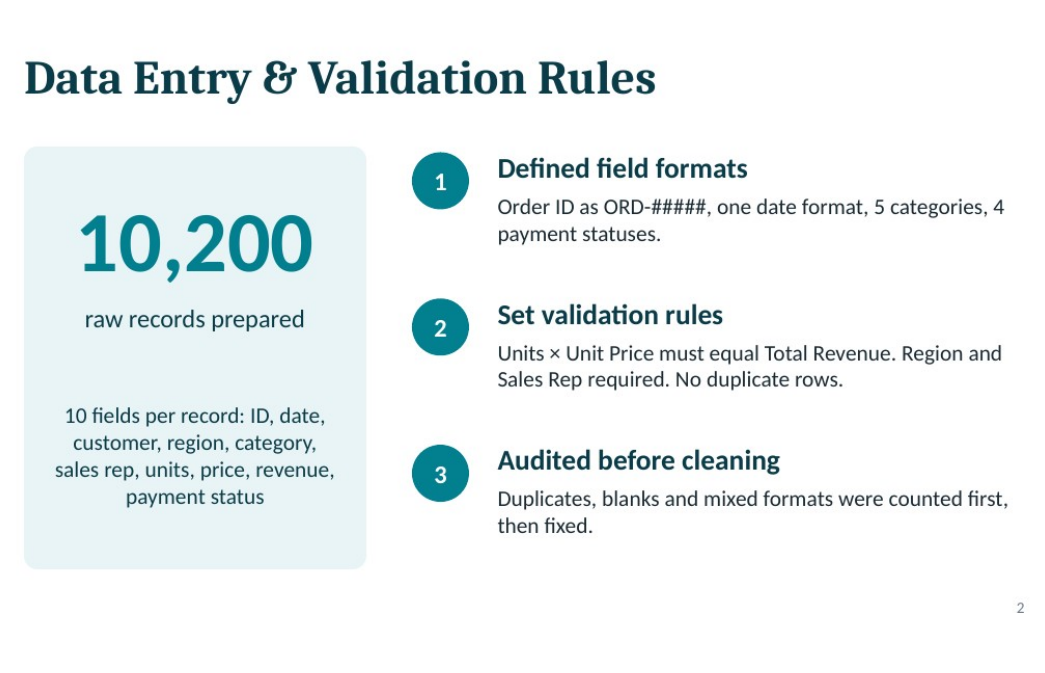
SELF-DIRECTED PROJECT

AI-Powered Data Management & MIS Reporting System

10,200 raw business records cleaned, validated and reported using Excel, AI and PowerPoint.

MS Excel
AI Assistance
MS PowerPoint

Based on a sample business dataset



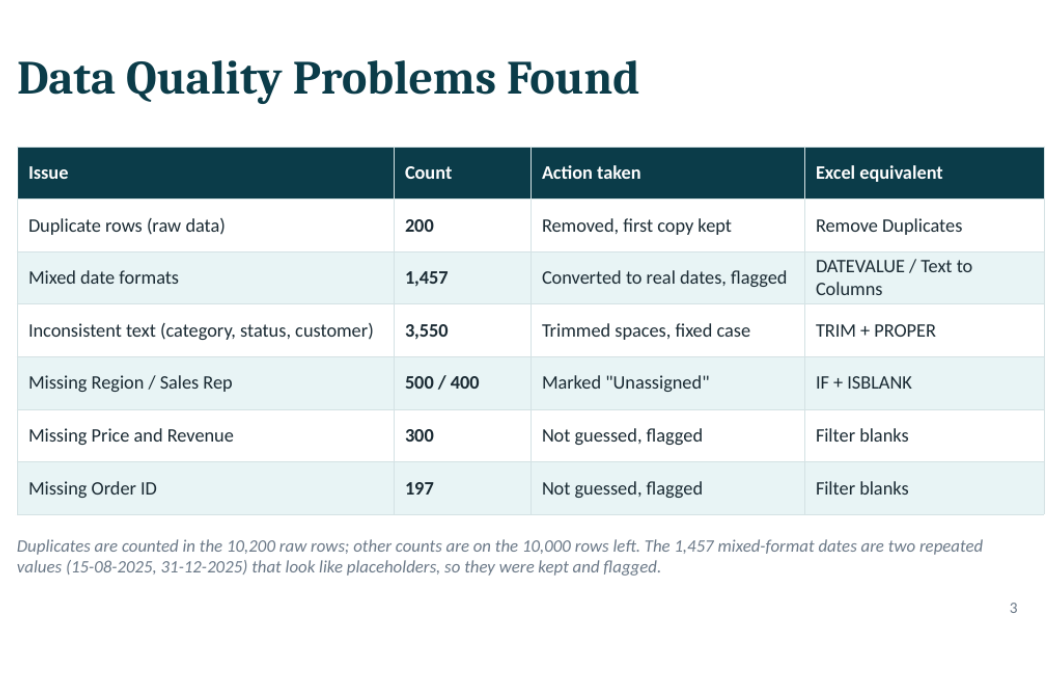
Data Entry & Validation Rules

10,200

raw records prepared

10 fields per record: ID, date, customer, region, category, sales rep, units, price, revenue, payment status

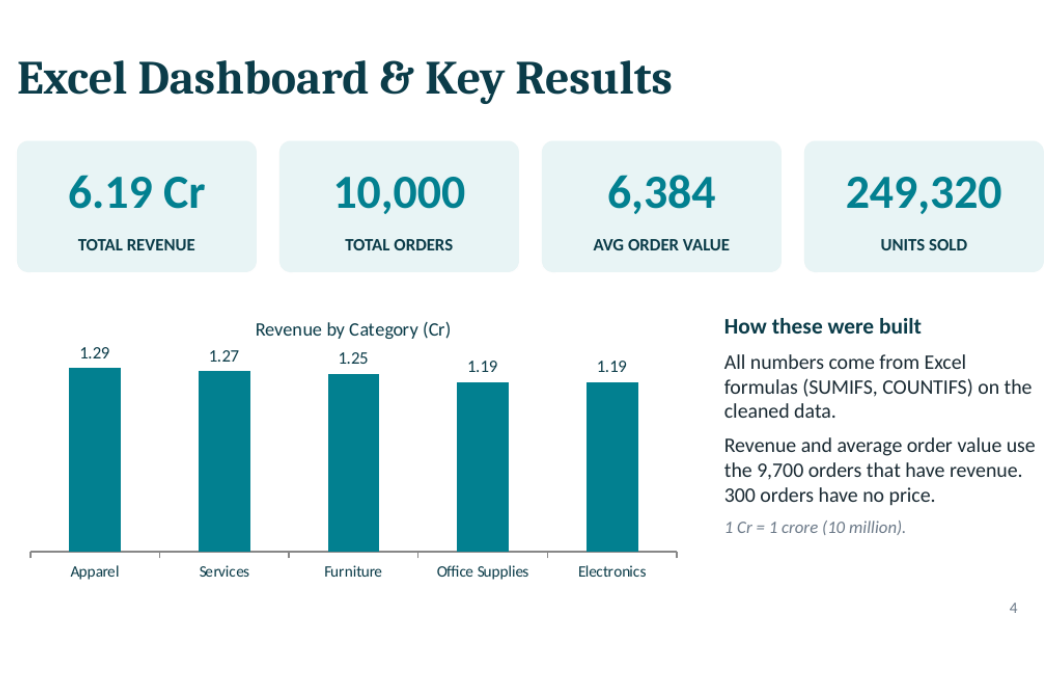
1. Defined field formats: Order ID as ORD-#####, one date format, 5 categories, 4 payment statuses.
2. Set validation rules: Units × Unit Price must equal Total Revenue. Region and Sales Rep required. No duplicate rows.
3. Audited before cleaning: Duplicates, blanks and mixed formats were counted first, then fixed.



Data Quality Problems Found

Issue	Count	Action taken	Excel equivalent
Duplicate rows (raw data)	200	Removed, first copy kept	Remove Duplicates
Mixed date formats	1,457	Converted to real dates, flagged	DATEVALUE / Text to Columns
Inconsistent text (category, status, customer)	3,550	Trimmed spaces, fixed case	TRIM + PROPER
Missing Region / Sales Rep	500 / 400	Marked "Unassigned"	IF + ISBLANK
Missing Price and Revenue	300	Not guessed, flagged	Filter blanks
Missing Order ID	197	Not guessed, flagged	Filter blanks

Duplicates are counted in the 10,200 raw rows; other counts are on the 10,000 rows left. The 1,457 mixed-format dates are two repeated values (15-08-2025, 31-12-2025) that look like placeholders, so they were kept and flagged.



Excel Dashboard & Key Results

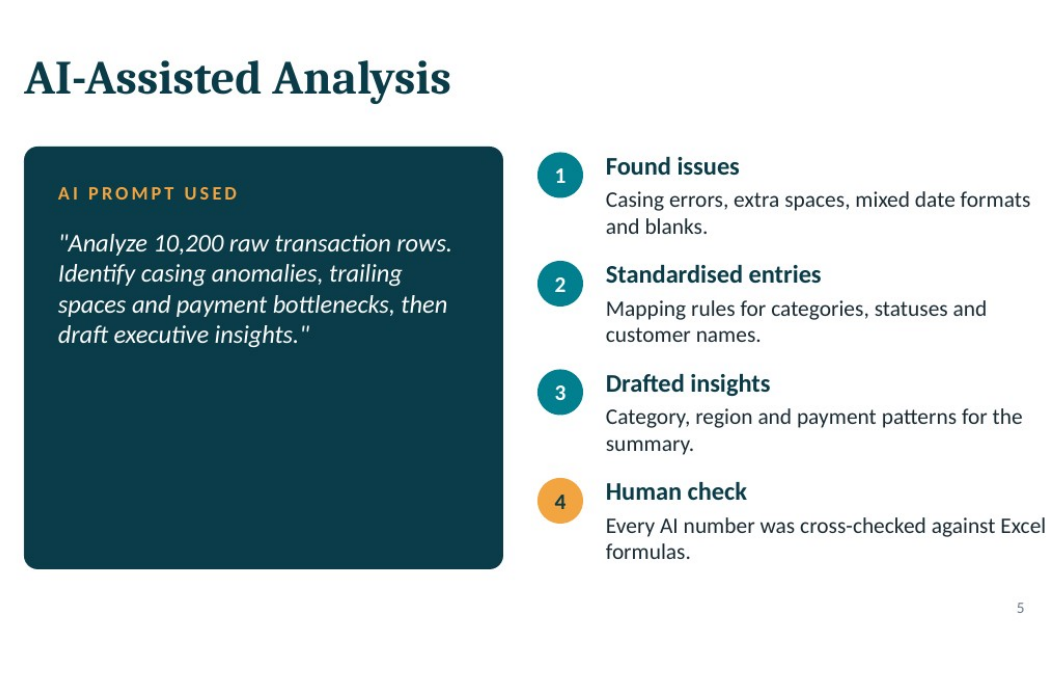
6.19 Cr	10,000	6,384	249,320
TOTAL REVENUE	TOTAL ORDERS	AVG ORDER VALUE	UNITS SOLD

How these were built

All numbers come from Excel formulas (SUMIFS, COUNTIFS) on the cleaned data.

Revenue and average order value use the 9,700 orders that have revenue. 300 orders have no price.

1 Cr = 1 crore (10 million).

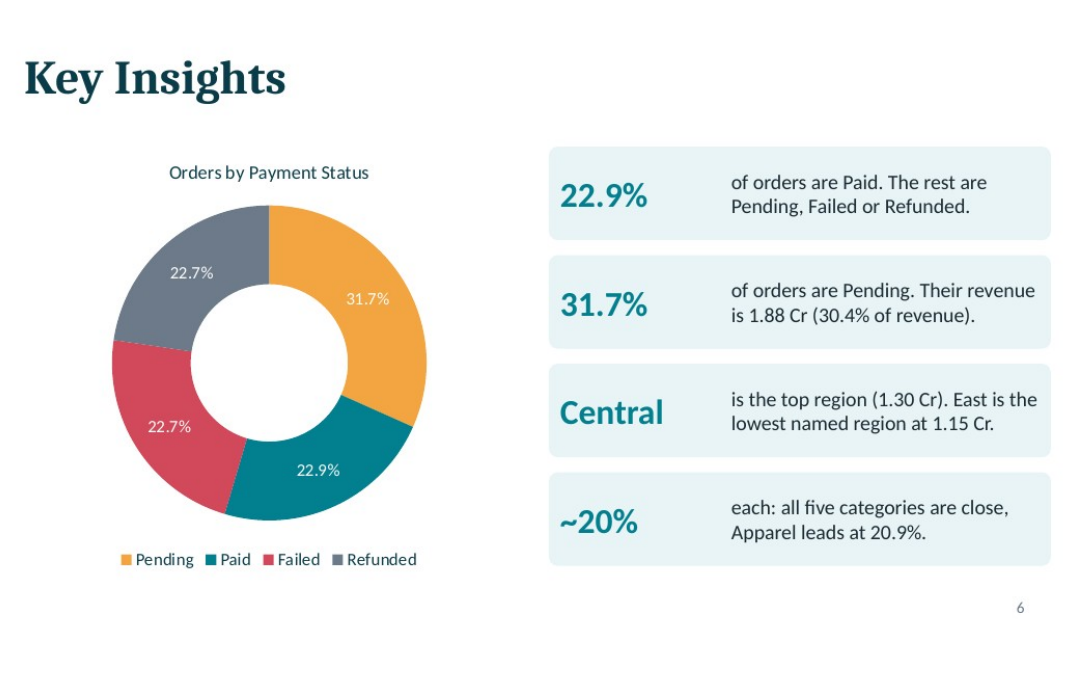


AI-Assisted Analysis

AI PROMPT USED

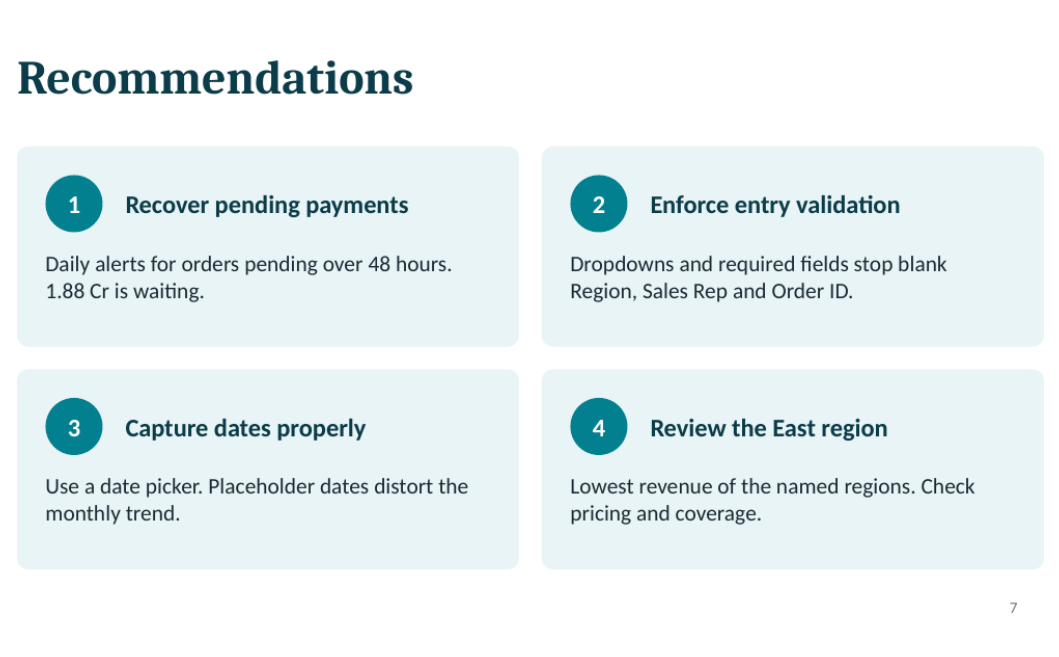
"Analyze 10,200 raw transaction rows. Identify casing anomalies, trailing spaces and payment bottlenecks, then draft executive insights."

1. Found issues: Casing errors, extra spaces, mixed date formats and blanks.
2. Standardised entries: Mapping rules for categories, statuses and customer names.
3. Drafted insights: Category, region and payment patterns for the summary.
4. Human check: Every AI number was cross-checked against Excel formulas.



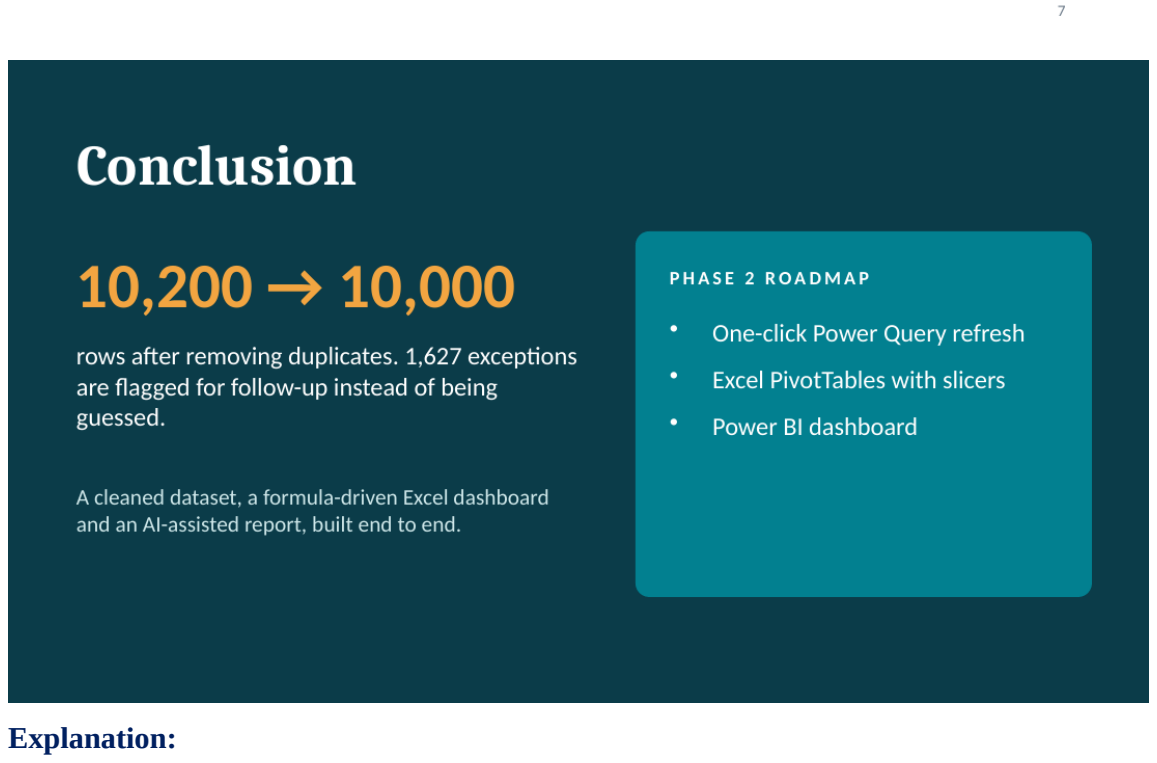
Key Insights

22.9%	of orders are Paid. The rest are Pending, Failed or Refunded.
31.7%	of orders are Pending. Their revenue is 1.88 Cr (30.4% of revenue).
Central	is the top region (1.30 Cr). East is the lowest named region at 1.15 Cr.
~20%	each: all five categories are close, Apparel leads at 20.9%.



Recommendations

1. Recover pending payments: Daily alerts for orders pending over 48 hours. 1.88 Cr is waiting.
2. Enforce entry validation: Dropdowns and required fields stop blank Region, Sales Rep and Order ID.
3. Capture dates properly: Use a date picker. Placeholder dates distort the monthly trend.
4. Review the East region: Lowest revenue of the named regions. Check pricing and coverage.



Conclusion

10,200 → 10,000

rows after removing duplicates. 1,627 exceptions are flagged for follow-up instead of being guessed.

A cleaned dataset, a formula-driven Excel dashboard and an AI-assisted report, built end to end.

PHASE 2 ROADMAP

- One-click Power Query refresh
- Excel PivotTables with slicers
- Power BI dashboard

Explanation:

- **Slide 1, Title:** introduces the project, an AI-powered data management and MIS reporting system built with MS Excel, AI assistance and MS PowerPoint on a sample business dataset.
- **Slide 2, Data Entry & Validation Rules:** the project started with 10,200 raw records and 10 fields. It defined field formats, set validation rules (for example, Units × Price must equal Revenue) and audited the data before cleaning.
- **Slide 3, Data Quality Problems Found:** lists the problems and the fix for each: 200 duplicates removed, 1,457 mixed dates converted, 3,550 text issues fixed, 500 blank regions and 400 blank sales reps marked "Unassigned", and 300 missing prices and 197 missing order IDs flagged and not guessed.
- **Slide 4, Excel Dashboard & Key Results:** total revenue is 6.19 crore across 9,700 orders with revenue data (the dataset has 10,000 orders), with an average order value of 6,384 and 249,320 units sold. A chart shows revenue by category, and all numbers come from Excel formulas.
- **Slide 5, AI-Assisted Analysis:** shows the AI prompt used and what AI did: found issues, standardised entries and drafted insights. Every AI number was cross-checked against Excel.
- **Slide 6, Key Insights:** only 22.9% of orders are Paid and 31.7% are Pending (their revenue is 1.88 crore, 30.4% of revenue). Central is the top region, East is the lowest named region, and all five categories are close to 20% each.
- **Slide 7, Recommendations:** recover pending payments with daily alerts, enforce entry validation, capture dates with a date picker, and review the East region.
- **Slide 8, Conclusion:** 10,200 rows became a cleaned dataset of 10,000 rows, with 1,627 exceptions flagged for follow-up. Phase 2 plans are Power Query refresh, PivotTables with slicers and a Power BI dashboard.

PowerPoint File Link:

https://github.com/afifahkhatoon61-beep/my-certificates-projects/blob/main/AI-Powered%20Data%20Management%20Presentation.pptx

AI-ASSISTED ANALYSIS

AI was used as a helper, not as the source of truth. It helped to:

- find casing errors, extra spaces, mixed date formats and blanks;
- suggest standard names for categories, statuses and customers;
- draft insights and a management summary.

Every AI-generated number was cross-checked against Excel formulas before it was used in the report.

KEY INSIGHTS

- Only 22.9% of orders are Paid. About 31.7% of orders are Pending, and their revenue is 1.88 crore (30.4% of revenue).
- Central is the top region by revenue (1.30 crore). East is the lowest of the named regions (1.15 crore).
- All five categories are close to 20% of revenue each. Apparel leads with 20.9%.
- 500 orders have no region and 400 have no sales rep, which shows weak data entry controls.

RECOMMENDATIONS

1. **Recover pending payments:** send daily alerts for orders pending more than 48 hours.
2. **Enforce entry validation:** use dropdowns and required fields so Region, Sales Rep and Order ID are never blank.
3. **Capture dates properly:** use a date picker, because placeholder dates distort the monthly trend.
4. **Review the East region:** check pricing and coverage, as it has the lowest revenue.

LIMITATIONS

- The data is sample data, so the findings show the method and not real business performance.
- 300 orders have no price or revenue, so revenue totals cover 9,700 orders.
- Two dates (15-08-2025 and 31-12-2025) repeat on 1,457 orders and look like placeholders. The August and December revenue spikes are linked to them and should not be taken as real trends.
- The summary tables are formula-based, pivot-style tables, not native Excel PivotTables.

SKILLS DEMONSTRATED

- Data formatting and data entry rules
- Data cleaning and validation
- Duplicate and missing-data detection
- Excel formulas, summary tables and dashboards
- Responsible use of AI for analysis
- MIS reporting and PowerPoint presentation

CONCLUSION

This self-directed project turned 10,200 messy records into a cleaned dataset of 10,000 rows (with 1,627 exceptions flagged for follow-up), an Excel dashboard and a management presentation. It shows the complete data management to MIS reporting workflow, and it is honest about what could not be fixed. As future work, I plan to add Power Query refresh, native PivotTables with slicers and a Power BI dashboard.